Bounded-support random bridges for mask-domain speech enhancement

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Abstract

Generative speech enhancement generalises better than discriminative enhancement to unseen acoustic conditions, but is held back by two documented costs: it needs tens of network evaluations per utterance, and it can hallucinate content that was not present in the input. We observe that both costs follow from a modelling choice rather than from generative modelling itself. Every diffusion and bridge model currently used for speech takes a Gaussian driving process, whose state space is all of $\mathbb{R}^n$; the quantity being estimated, a time–frequency mask, lives on $[0, 1]$. We construct a random bridge driven by a Wright–Fisher process, whose state space is $[0, 1]$ by construction, and derive its transition kernel as an exact Doob h-transform of a finite binomial chain, avoiding the series expansion the Wright–Fisher diffusion would otherwise require. On the empirical distribution of ideal ratio masks the resulting bridge attains a given transport accuracy in fewer sampling steps than a Gaussian bridge, by a factor of 2.3 at five steps. Because a mask in $[0, 1]$ can only attenuate the observed signal, a mask-domain generative model cannot synthesise content absent from its input, so the hallucination failure mode is excluded by the construction rather than mitigated after the fact. *Sections 7 onward are a plan; results reported here are one-dimensional and use exact conditional expectations.*

1 Introduction

Discriminative speech enhancement maps a noisy observation to a clean estimate directly. Generative enhancement instead models the distribution of clean speech and conditions on the observation. The generative route is more robust when test conditions differ from training conditions [2, 3], which is the reason to pay for it, but two costs are consistently reported. Diffusion-based systems require tens of network evaluations per utterance, placing them one to two orders of magnitude above discriminative systems in arithmetic cost [4]. They can also insert or substitute phonetic content, produce spurious breathing, and introduce tonal artefacts; these hallucinations are described in the literature as an open problem [4, 5].

Both costs are usually treated as intrinsic to generative modelling. We argue they follow instead from the choice of driving process. Existing diffusion and bridge formulations for speech — score-based models, Schrödinger bridges and their one-step variants — all drive the transport with Brownian motion, whose state space is $\mathbb{R}^n$. When the estimated quantity is a time–frequency mask, the state space is $[0, 1]$, and a Gaussian transport spends part of every trajectory outside it.

Contributions.

1. We measure how much of a Gaussian bridge's trajectory leaves the feasible set on real speech, for three representations, and show the violation is substantial for masks and absent for the log-magnitude representation that existing systems use (Section 3).

2. We separate two properties that a support-matched driver may carry, boundedness and monotonicity, and show by ablation that monotonicity is the expensive one. This rules out the subordinator families and identifies the requirement: bounded and non-monotone (Section 4).

3. We construct the Wright–Fisher bridge, deriving its transition kernel as an exact h-transform of a finite binomial chain and so avoiding the infinite mixture representation the diffusion limit requires (Section 5).

4. We show on the empirical mask distribution that the Wright–Fisher bridge reaches a given accuracy in fewer sampling steps than a Gaussian bridge, and explain the mechanism (Section 6).

5. We set out why a mask-domain generative model cannot hallucinate, and specify the enhancement experiment that would test both claims (Section 7).

2 Background

2.1 Random bridges

We use the formulation of [1]. For a càdlàg driving process $\{Z_t\}_{t\in[0,T]}$ and laws Φ, Ψ with $\Phi \ll \mu_{Z_0}$ and $\Psi \ll \mu_{Z_T}$, a process $\{\xi_t\}$ is a (Φ, Ψ)-bridge under Z if its endpoints are distributed as a coupling of Φ and Ψ and, conditional on those endpoints, its finite-dimensional distributions agree with those of Z conditioned likewise. The network estimates the L^2 projection $f_\theta(\xi_s, s) \approx \mathbb{E}[Y \mid \xi_s]$ under squared error, and generation integrates a transport whose terminal law is Ψ.

The definition constrains the driver only to be càdlàg. Its Gaussian specialisation recovers diffusion bridges; the present paper uses a driver that the definition admits but that no existing speech system employs.

2.2 The Gaussian case is what current systems implement

With $Z_t = \sigma W_t$ and $t = sT$, writing $\gamma^2 = \sigma^2 T$, the bridge marginal conditional on the endpoints is

$$\xi_s = (1-s)x + sy + \gamma\sqrt{s(1-s)}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{1}$$

and the exact transition from s to u pinned at an endpoint estimate $\hat{y}$ is Gaussian with mean $\xi_s + \frac{u-s}{1-s}(\hat{y} - \xi_s)$ and variance $\gamma^2 \frac{(u-s)(1-u)}{1-s}$. Score-based enhancement, diffusion bridges and Schrödinger bridges for speech differ in their schedule, parameterisation and training objective, but all place the state in $\mathbb{R}^n$ and all inherit (1)'s unbounded support.

3 The support violation is real, and specific to masks

Measurement. We take real speech, add noise at 5 dB, and form three representations of the enhancement problem: the linear magnitude spectrogram, with support $[0, \infty)$; the ideal ratio mask, with support $[0, 1]$; and the phase, which is circular. For each we construct the Gaussian bridge (1) between the true endpoints and record the fraction of path mass outside the feasible set.

Why the mask is the case that matters. The magnitude constraint has an escape: taking logarithms removes it, which is what existing systems do, and the control row of Table 1 confirms the escape works. A mask admits no such transformation. It is bounded on both sides, and the standard device — a sigmoid on the network output — is a projection applied after the fact, not a process that respects the constraint.

Table 1: Fraction of a Gaussian bridge's trajectory outside the feasible set, on 149 s of real speech. Excess is relative to each representation's own scale. The log-magnitude row is a control: its support is unconstrained, and the measurement returns zero.

Representation	Support	Outside	Mean excess
Linear magnitude	$[0, \infty)$	11.1%	0.074
Ratio mask	$[0, 1]$	8.0%	0.063
Phase	circular	1.3%	0.069
Log magnitude (control)	$\mathbb{R}$	0.00%	0

The mask distribution also concentrates at the boundary: 81% of time–frequency bins fall in the lowest decile and 77% of mass lies within 0.05 of 0 or 1. A transport that overshoots does so into the region where the target places most of its mass.

4 Which property of the driver matters

A driver matched to $[0, 1]$ may be bounded, monotone, or both. The gamma and Poisson bridges derived in [1] are both: they are driven by subordinators, so their paths cannot decrease. We separate the two properties by ablation, taking a Gaussian bridge and forbidding its steps to decrease while leaving the driver, the noise and the support handling unchanged.

Table 2: Monotonicity is the expensive property. Targets are Gamma laws of varying shape at fixed mean, so only the behaviour of the density at the boundary changes; 1-Wasserstein distance at 50 steps, exact conditional means.

Shape k	Gaussian	Gaussian, monotone	Gamma bridge	monotone / free
0.5	0.106	1.412	0.362	13.3×
2.0	0.033	0.498	0.083	15.1×
8.0	0.012	0.102	0.019	8.3×

Forbidding decrease costs between 8 and 15 times, against the factor of two to three by which the gamma bridge trails a free Gaussian. The subordinator families are therefore the wrong choice for a quantity that is not ordered relative to its starting point, and a mask is not: a given bin's ideal mask may lie above or below any initial estimate. The requirement is a driver that is bounded and *not* monotone.

5 The Wright–Fisher bridge

5.1 The driver

The Wright–Fisher diffusion on $[0, 1]$,

$$\mathrm{d}X_t = \tfrac{1}{2}\big(\theta_1(1 - X_t) - \theta_2 X_t\big)\mathrm{d}t + \sqrt{X_t(1 - X_t)}\,\mathrm{d}W_t, \tag{2}$$

is the canonical diffusion with this state space. It is continuous, hence càdlàg, so it is admitted by the bridge definition; its diffusion coefficient is state dependent, so it has neither homogeneous nor independent increments and the Lévy results of [1] do not reach it. Its paths are not monotone.

With $\theta_1 = \theta_2 = 0$ both boundaries absorb, and the law of X_T carries atoms at 0 and 1 together with a density between. This matters for admissibility: the requirement $\Psi \ll \mu_{Z_T}$ fails for a mask distribution with boundary atoms if the boundaries are inaccessible, which is the case under strong mutation. The neutral process is the member of the family whose terminal law has the same atomic structure as the data.

5.2 Transition kernel without the series expansion

The transition density of the Wright–Fisher diffusion is available only as an infinite mixture over a death process [6, 7]. The finite population model of which it is the diffusion limit has no such difficulty. With population size M, one generation is a binomial resampling,

$$P(i \to j) = \binom{M}{j} \left(\tfrac{i}{M}\right)^{j} \left(1 - \tfrac{i}{M}\right)^{M-j}, \qquad i, j \in \{0, \dots, M\}, \tag{3}$$

an exact finite stochastic matrix, with 0 and M absorbing.

Proposition 1 (Wright–Fisher bridge kernel)**.** *Let $\{\xi\}$ be the bridge under the chain* (3) *pinned at $\xi_1 = y$, and let a and b be the generation counts on $[s, u]$ and $[u, 1]$. Then*

$$\mathbb{P}(\xi_u = j \mid \xi_s = i,\ \xi_1 = y) = \frac{P^a[i, j]\ P^b[j, y]}{P^{a+b}[i, y]}, \tag{4}$$

which is the Doob h-transform of (3) *with space–time harmonic function $h_u(j; y) = P^b[j, y]$.*

Proof. Immediate from the Markov property of the chain and Bayes' rule; the denominator is the normalisation $\sum_j P^a[i, j]P^b[j, y] = P^{a+b}[i, y]$. □

Every quantity is a power of a finite matrix. For $M = 128$ and a few hundred generations the powers are precomputed once.

Remark 2 (Absorbed states)**.** *Because both boundaries absorb, a path that has reached* 0 *or* 1 *cannot reach a pin that disagrees with it, and the kernel of Proposition 1 is undefined there. Such paths remain where they are, which is what absorption means. In sampling this arises when the endpoint estimate moves after a trajectory has absorbed.*

5.3 Sampling

Training is unchanged from the Gaussian case: sample a bridge time, sample the state from the marginal, regress the endpoint under squared error. Generation alternates evaluating f_θ and sampling Proposition 1 with the estimate in place of the unknown endpoint, exactly as ancestral diffusion sampling substitutes a point estimate for the posterior.

6 Why the bounded driver needs fewer steps

Table 3: Transport accuracy against sampling steps on the empirical distribution of ideal ratio masks. 1-Wasserstein distance, exact conditional means, $M = 128$, 256 generations, 5% relative error injected into the endpoint estimate. Floor 0.0027.

Steps	2	5	10	20	50
Gaussian, projected at the end	0.0715	0.0485	0.0335	0.0253	**0.0179**
Gaussian, projected each step	0.0715	0.0487	0.0341	0.0262	0.0198
Wright–Fisher	**0.0561**	**0.0208**	**0.0192**	**0.0200**	0.0207
Ratio	0.79	0.43	0.57	0.79	1.15

The Wright–Fisher bridge is ahead below fifty steps and by a factor of 2.3 at five. The two cross at fifty.

Mechanism. A Gaussian transport towards a boundary-concentrated target overshoots, and correcting an overshoot costs steps: the drift $\frac{\hat{y}-\xi_s}{1-s}$ must first reverse the excursion and then re-approach. Given enough steps the correction completes, which is why the Gaussian catches up at fifty. The Wright–Fisher transport cannot overshoot, because its diffusion coefficient $\sqrt{x(1-x)}$ vanishes at both boundaries, so the steps it does take are not spent undoing earlier ones. The advantage is therefore largest where the step budget is smallest, which is the regime in which generative enhancement is currently impractical.

The two crossing at fifty steps is consistent with this account rather than a contradiction of it: an unconstrained transport with a sufficient budget is the more flexible of the two, and the constraint stops paying once the budget is ample.

7 Application: mask-domain enhancement

7.1 Why the mask domain excludes hallucination

Let $\hat{S} = m \odot X$ with $m \in [0, 1]$ and X the observed noisy spectrogram. Then $|\hat{S}_{tf}| \leq |X_{tf}|$ for every bin, so the estimate is supported on the observation. A mask-domain generative model can suppress and it can fail to suppress, but it cannot place energy where the input had none. The phonetic insertions and spurious vocalisations reported for waveform and complex-spectrum diffusion [4, 5] are not available to it.

This is a property of the output parameterisation rather than of the driver. What the driver supplies is the ability to model a distribution over masks without leaving $[0, 1]$, so that the guarantee is not obtained by projecting a model that does not respect it.

7.2 Planned experiment

To be run. Enhancement on VoiceBank–DEMAND with a mismatched evaluation set, since mismatch is where the generative route earns its cost. The transport runs from a cheap deterministic mask computed from the observation to the ideal ratio mask, both conditioned on the same noisy spectrogram; this uses the bridge's arbitrary source distribution rather than starting from noise.

Arms: a discriminative mask predictor; a Gaussian mask bridge, which is the ablation isolating the driver; a Wright–Fisher mask bridge; and a published waveform or complex-spectrum diffusion system, which is where hallucination has been reported. All share architecture, budget and evaluation code.

Measures: PESQ, ESTOI and SI-SDR for quality; network evaluations and wall-clock for cost; and, for hallucination, word and phoneme error rate under a fixed ASR system, which is the measure on which mask-domain models should be unimprovable-upon by construction and waveform models should not.

7.3 What would falsify the claim

If a Gaussian mask bridge matches the Wright–Fisher bridge at five to ten steps on real data, the mechanism of Section 6 does not survive contact with a learned conditional expectation, and the contribution reduces to the derivation. If mask-domain enhancement cannot reach the quality of complex-spectrum systems at any step count, the hallucination guarantee is purchased at a price the application will not pay.

8 Limitations

The results in Tables 2 and 3 are one-dimensional, use exact conditional expectations rather than a trained network, and treat time–frequency bins independently. They establish a mechanism

and its size in a setting where nothing else can be responsible; they do not establish that the mechanism survives at spectrogram scale with a learned estimator. The speech measurements of Table 1 use a small corpus and synthetic additive noise. No enhancement experiment has been run.

References

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